An Introduction to Quantum Machine Learning (QML)

Fundamentals, Algorithms, and the Reality of NISQ Devices

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March 2026

Abstract

This document presents a technical analysis of quantum computing paradigms applied to machine learning (QML) and optimization. In the current **NISQ** (Noisy Intermediate-Scale Quantum) era, we explore the distinction between purely quantum (*fault-tolerant*) algorithms, such as Shor's and Grover's, and hybrid variational algorithms, such as **VQE** and **QAOA**, which use parameterized circuits and classical optimization. We also investigate classical-quantum approaches and dissipative models, which use interaction with thermal reservoirs for data classification. We discuss encoding methods (*feature maps*) and the use of quantum kernels to express problems in high-dimensional Hilbert spaces. Finally, we detail the challenges of trainability, including the *Barren Plateau* phenomenon, and the analytical gradient calculation methods via the *parameter-shift rule*. The objective is to provide a robust theoretical foundation for understanding quantum advantage and its state-of-the-art applications.

1 Introduction

Current quantum computing resides in the **NISQ** (Noisy Intermediate-Scale Quantum) era, where devices have tens to hundreds of qubits with considerable noise and a lack of large-scale error correction. The goal of Quantum Machine Learning (QML) is to explore the fundamental limits of data analysis allowed by the laws of physics, using properties such as superposition and entanglement.

2 Taxonomy of Quantum Algorithms

The classification of quantum algorithms can be made based on the type of hardware required and the role of classical computing. In this section, we distinguish three major categories: purely quantum (*fault-tolerant*) algorithms, hybrid (variational) algorithms, and classical-quantum approaches.

2.1 Purely Quantum Algorithms (Fault-Tolerant)

These algorithms are designed for ideal quantum computers with error correction (QEC). They focus on exact mathematical subroutines and offer guarantees of asymptotic speedup over the best known classical algorithms. Their main challenge is the need for thousands or millions of physical qubits to implement error correction, thus being beyond the reach of current NISQ devices.

- **Shor's Algorithm:** Solves the integer factorization problem in polynomial time, threatening encryption systems such as RSA. The speedup is exponential, since the best known classical algorithm is subexponential.

- **Grover's Algorithm:** Performs searches in unstructured databases with quadratic speedup ($O(\sqrt{N})$), reducing the number of queries to an oracle. Although more modest than Shor's, it is one of the bases of post-quantum cryptography.
- **HHL Algorithm:** Solves systems of linear equations $Ax = b$ in logarithmic time in the number of variables, serving as a basis for quantum linear regression, SVMs, and other learning routines.

These algorithms share the need for a fault-tolerant quantum computer, with long coherence times and precise operations. They represent the long-term future of quantum computing.

2.2 Hybrid Algorithms (Variational)

Variational algorithms are the main strategy for the NISQ era. They use an *ansatz* (parameterized circuit) that runs on quantum hardware, while a classical optimizer adjusts the parameters to minimize a cost function. In this way, the quantum part is kept short (low circuit depth) and the classical part compensates for the lack of error correction.

- **VQE (Variational Quantum Eigensolver):** Originally developed for quantum chemistry, it estimates the ground state of molecular Hamiltonians. The ansatz can be the *Unitary Coupled Cluster* (UCC) or *Hardware-Efficient Ansatz* (HEA), and the classical optimizer minimizes the energy.
- **QAOA (Quantum Approximate Optimization Algorithm):** Inspired by adiabatic evolution, it solves binary combinatorial optimization problems. It alternately applies problem and mixer operators, parameterized by angles γ and β. The number of layers p controls the solution quality.
- **Variational Classifiers:** They map classical data into quantum states via *feature maps* and use parameterized measurements to separate classes. Examples include variational circuits for binary and multi-class classification.

Common elements among these models: all use a parameterized circuit, a classical optimizer, and the cost function is obtained through repeated measurements. The difference lies in the ansatz structure and the nature of the problem (ground state vs. combinatorial optimization vs. classification).

2.3 Classical-Quantum and Dissipative Algorithms

This category includes algorithms that do not require full quantum hardware but are inspired by quantum principles to improve classical performance, or that use the dynamics of open quantum systems for learning tasks.

- **QWOA (Quantum Whale Optimization Algorithm):** It is a classical metaheuristic optimization algorithm that represents the population of solutions via simulated Q-bit amplitudes. Each attribute is associated with a Q-bit (α, β), and population evolution is performed by simulated rotations inspired by quantum gates. Despite using quantum terminology, it does not require quantum hardware, being executed entirely on classical computers.
- **Thermal Classifiers:** They use a two-level system (qubit) in contact with thermal reservoirs. The effective temperature in the steady state, determined by the interaction with data, serves as a classification criterion. These models explore the dynamics of open systems and can be implemented in current quantum devices or in simulations.
- **Quantum Boltzmann Machines:** Quantum extensions of Ising models, where the quantum Gibbs state is sampled for probabilistic learning. They can be trained on quantum hardware or simulated classically.

The main difference from hybrid algorithms is that in classical-quantum ones, quantum computing may not be present (as in QWOA) or is used more specifically (as in thermal classifiers). Hybrids, on the other hand, necessarily require a quantum processor to run the parameterized

circuits.

3 Data Encoding and Feature Maps

The encoding step maps classical data to quantum states, defining the *feature map* and the implicit kernel. The choice of encoding directly influences the expressivity of the model.

- **Basis Encoding:** Each sample is represented by a computational basis state (e.g., bit-strings). It is efficient in number of qubits but requires many circuits to represent continuous data.
- **Amplitude Encoding:** Associates data with probability amplitudes, allowing N dimensions to be represented with $\log N$ qubits. It is very economical but requires deep circuits to prepare the state.
- **Angle/Rotation Encoding:** Uses rotation gates (R_X, R_Y, R_Z) to encode each feature as a rotation angle. It is simple and suitable for variational circuits, but the number of qubits scales linearly with dimensionality.

The concept of *quantum kernel* arises from the equivalence between a quantum classifier and a classical SVM with kernel defined by inner products between quantum feature maps. Quantum kernels can generate high-dimensional feature spaces inaccessible classically.

4 Optimizers and Training

Training variational models is often an **NP-hard** problem due to the non-convex nature of the cost surface. The optimizers used are divided into two major classes:

4.1 Gradient-Based Optimizers

These use the *parameter-shift rule* to compute exact gradients on hardware. The rule states that for gates of the form $e^{-i\theta G}$ with $G^2 = I$, the derivative can be obtained by the difference of two circuit evaluations shifted by $\pi/2$.

- **SGD (Stochastic Gradient Descent):** Updates parameters using a gradient estimate, suitable for large datasets.
- **Adam:** Adapts the learning rate per parameter, combining momentum and bias correction.
- **SPSA (Simultaneous Perturbation Stochastic Approximation):** Uses random perturbations to estimate the gradient, being more robust to measurement noise.

4.2 Gradient-Free Optimizers

Used when the cost of estimating gradients is prohibitive or the noise is too high.

- **COBYLA (Constrained Optimization BY Linear Approximations):** Builds linear approximations of the objective function and minimizes within a trust region. It is widely used in VQE and QAOA implementations.
- **Nelder-Mead:** Direct search method based on a simplex, does not require derivatives.
- **CMA-ES (Covariance Matrix Adaptation Evolution Strategy):** Robust evolutionary algorithm for non-convex problems.

A common element to all optimizers is the need for estimates of the cost function (or gradient) from finite samples (*shot noise*), which introduces statistical noise and can hinder convergence.

5 Main Challenges

Despite progress, quantum machine learning faces fundamental obstacles:

- **Barren Plateaus:** In very deep or expressive ansatzes, the gradient variance decays exponentially with the number of qubits. This makes training impractical, as the average gradient becomes indistinguishable from noise.
- **Input Problem:** Loading classical data into quantum states can have exponential cost (e.g., amplitude encoding requires deep circuits). If state preparation is more expensive than the algorithmic gain, quantum advantage is lost.
- **Noise and Decoherence:** Real qubits suffer interactions with the environment (decoherence), and quantum gates have limited fidelity. This destroys the coherence required for many algorithms.
- **Shot Noise:** The finite number of measurements introduces statistical error in the estimation of expected values, requiring a trade-off between precision and execution time.

6 Applications and State of the Art

QML has shown potential in several domains, especially when combined with quantum kernels and error mitigation techniques.

- **Healthcare:** Diagnosis via image segmentation (e.g., QFS-Net for brain MRI) and drug discovery. Quantum kernels have been applied to genomic data classification with superior performance to classical methods.
- **Finance:** Portfolio optimization, risk modeling, and fraud detection. Algorithms like QAOA have been explored for portfolio optimization problems.
- **Industry:** Traffic optimization, logistics, and new material design (via VQE).
- **Quantum Kernels:** SVMs with quantum kernels enable multi-class classification in high-dimensional Hilbert spaces, surpassing the expressivity of classical kernels like RBF.
- **Quantum System Simulation:** VQE and its derivatives are used to study properties of molecules and materials, with the potential to revolutionize computational chemistry.

Although there is not yet proven quantum advantage in practical machine learning problems, rapid hardware evolution and the development of error mitigation methods are paving the way for useful applications in the coming years.

7 Conclusion

The integration between artificial intelligence and quantum computing is constantly evolving. The success of the field depends on the creation of standardized quantum *datasets*, the development of error mitigation methods that enable the transition from NISQ utility to true quantum advantage, and a clear understanding of the differences between purely quantum, hybrid, and classical-quantum approaches. Each of them plays a role in the current and future ecosystem, and the appropriate choice depends on the problem, hardware resources, and tolerance to noise.

8 Useful Links

- https://quantum.cloud.ibm.com/learning/en/courses/;
- https://pennylane.ai/qml

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